KAVACH and NIRMAN Website content brief

SAG Aerospace and Defence | Product content and visual reference

We are developing indigenous defence systems that combine unmanned platforms, sensing and deployable manufacturing. This brief provides the product descriptions, variants, capability highlights and visual references needed to build the KAVACH and NIRMAN sections of our website.

Product introductions

SAG KAVACH is a proposed layered counter-drone protection system for military transport convoys and high-value assets. It brings sensor cueing, electro-optical tracking, optical camera disruption and physical net capture into a common protection architecture.

SAG NIRMAN is a proposed containerised drone factory and spares workshop. It brings drone production, repair and scan-to-part manufacturing closer to the point of need through Army, Air Force and Navy configurations.

Product page	Primary message
KAVACH	Layered counter-drone protection for vehicles and sites
NIRMAN Army	Forward drone production and field equipment spares
NIRMAN Air Force	Air-mobile production for base defence, training and GSE support
NIRMAN Navy	Drone replenishment and ship spares at sea or ashore

SAG KAVACH

Layered counter-drone protection for convoys and high-value assets



KAVACH is designed to help protect transport vehicles, convoy escorts, radars, launchers and command posts against small drones. External cueing from SAG PRAHARI supports day and thermal tracking; a multi-band dazzler targets drone cameras, while a physical net-capture layer addresses persisting threats. Acoustic and optical functions are intended to address RF-silent and fibre-optic drones within their respective sensing limits.

Three configurations

KAVACH-C Convoy Node: Full sensor-cued tracking, dazzler and net-launcher configuration for lead, tail or escort vehicles, with convoy track sharing.
KAVACH-T Truck Kit: Compact EO and ranging package with a net launcher, linked to convoy-node tracks for transport-truck retrofit.
KAVACH-S Site Defence: Mast or pedestal installation for local protection of air-defence and other high-value static assets.

Capability	Proposed target
EO detection	DJI-class quad: ≥2 km day / ≥1.5 km thermal
Camera disruption in clear air	ISR drones: up to 1 km; FPV drones: 300–500 m
Net capture envelope	30–150 m nominal; up to 200 m for slow or hovering targets
Launcher and power	12 cartridges per launcher; 24 V DC; ≤600 W peak

Operator arming, no-fire sectors, range-based controls and safety interlocks are planned. Performance depends on target sensors, weather and visibility; laser safety and system effectiveness require validation. Existing passive protection remains the final layer.

SAG NIRMAN Army

Forward-deployed mobile drone factory and field spares workshop



The Army variant is designed to reduce reliance on long supply lines for small drones and replacement parts. Two mobile modules bring production and repair to forward formations, supporting dispersed sites and rapid changes to drone designs.

Configuration and applications

Module A combines drone airframe printing, foam cutting, kit assembly and test facilities. Damaged drone structures can be reprinted and serviceable electronics reused. Module B adds scanning, CAD/CAM, metal deposition, CNC finishing, heat treatment and inspection for field spares.

Intended users include forward formations and EME workshops. Typical spares include brackets, handles, covers, mounts and bushings for vehicles and legacy equipment. Initial production focuses on drone structures and non-critical spares; safety-critical parts require qualification and user approval.

Army capability	Proposed target
Deployment	2 × 20 ft ISO modules; ≤10 t each
Module dimensions	6.06 × 2.44 × 2.59 m each
Movement and power	Truck / hook-lift, rail, planned airlift; 62.5 kVA generator or grid
Production across two shifts	≥30 FPV / ISR quadcopters OR ≥6 fixed-wing ISR drones per day
Readiness	First job ≤2 hr after arrival; packing ≤1 hr
Field support	≥200 drone kits; 6 personnel per shift; 3-week training
Environment	−20°C to +50°C; up to 4,500 m with derating

SAG NIRMAN Air Force

Air-mobile drone factory and spares workshop for forward air bases



The Air Force variant is designed to produce base-defence and ISR drones, low-cost aerial targets and ground-support spares at air stations and forward bases. It aims to reduce the need to airlift finished drones and wait for small maintenance items.

Configuration and applications

Module A supports interceptor and ISR quadcopters, fixed-wing ISR aircraft and aerial targets, using printed structures, foam surfaces and stocked electronics kits. Target designs provide for visual or radar augmentation and scoring aids. Module B supports GSE parts, tooling, fixtures, brackets, protective caps, covers and servicing adaptors.

The initial scope excludes aircraft parts. Any future airworthy part requires CEMILAC and DGAQA qualification. Interceptor airframes are issued without energetics or effectors; their fitment follows service procedures.

Air Force capability	Proposed target
Deployment	2 × air-mobile 20 ft modules; ≤9 t each
Module dimensions	6.06 × 2.44 m footprint; 2.44 m height
Transport design intent	C-17, IL-76 and C-130J; road movement from airhead
Production across two shifts	≥30 interceptor / ISR quadcopters OR ≥6 fixed-wing ISR / target drones per day
Readiness	First job ≤2 hr from drive-off; packing ≤1 hr
Power and stock	62.5 kVA generator or station supply; ≥200 drone kits
Environment	−10°C to +50°C; coastal and desert base operation

SAG NIRMAN Navy

Ship-borne drone factory and spares workshop for ship and shore



The Navy variant is designed to replenish drones and manufacture replacement parts while ships remain on station. It also supports island territories and forward naval bases, with modules that can be craned ashore.

Three module ship set

Module A produces ISR, interceptor and gunnery-target drones. Module B provides scanning, metal and polymer manufacturing, finishing and inspection for ship spares. Module C carries drone kits and feedstock, protected battery storage, the ship-power interface, command console and secure parts-library server.

Marine material targets include 316L and duplex stainless steel, nickel-aluminium bronze and Inconel 625. Example parts include valve handwheels, pump impellers, brackets, flanges, covers, cable glands, gaskets and seals, subject to qualification and approval.

Navy capability	Proposed target
Deployment	3 × marinised 20 ft ISO modules; ≤10 t each
Production across two shifts	≥30 ISR / interceptor quadcopters OR ≥6 small fixed-wing ISR / target drones per day
At-sea production	Up to sea state 5; secure-for-sea mode above
Readiness and stock	First job ≤2 hr after lashing and power connection; ≥300 drone kits
Ship power	About 65 kVA; 415 / 440 V, 50 / 60 Hz, 3-phase via Module C
Crew and environment	5 per shift; 3-week training; 0°C to +50°C ambient

Marinisation includes corrosion protection, sealed openings, equipment isolation, climate control and shipboard safety provisions. Deck suitability, motion performance and ship interfaces require vessel-specific assessment and trials.